Project Planning & Management Document

Project: MobiGuide (Market Driven Phone Recommender)

1. Project Proposal

Executive Overview

MobiGuide is a data analytics and recommendation solution for the Egyptian smartphone e-commerce market. Consumers can find it difficult to compare smartphones because prices, specifications, and listings differ across Amazon Egypt, Noon Egypt, and Jumia Egypt. For this project, product data was collected through a one-time web-scraping process and saved as a static dataset. The dataset is cleaned and standardized, custom performance metrics are calculated, and an interactive six-page Power BI dashboard helps users compare available options according to budget and usage priorities. Because the collection was performed once, the dashboard reflects the collected snapshot and does not automatically retrieve new listings or current prices.

Objectives

- Market Overview: Analyze the collected smartphone listings from Amazon Egypt, Noon Egypt, and Jumia Egypt to summarize the brands, models, prices, and platform coverage represented in the dataset.
- Cross-Platform Price Comparison: Match the same smartphone model across platforms where the available fields support a reliable match, then highlight the lowest recorded price in the collected dataset (Cheapest_Platform_Flag).
- Intelligent Budget Segmentation: Classify devices into four structured price tiers to simplify consumer browsing.
- Persona-Based Recommendation: Formulate mathematical scoring matrices (Gaming, Camera, Battery, Value-for-Money) that map specific user priorities (Personas) to top-performing devices within any budget.

Project Scope

- One-time web scraping from Amazon Egypt, Noon Egypt, and Jumia Egypt, with the collected results saved as a static dataset.
- Data cleaning, missing value handling, rating text extraction, and field standardization.
- Multi-page Power BI Dashboard development

- Quality assurance, user interaction testing, insight extraction, and executive presentation deck creation.

Data currency limitation: All analysis is based on the one-time scrape available for this project. Prices and availability may have changed since collection; the dashboard should not be described as showing live market prices.

2. Task Assignment & Team Roles

1. Omar Adel : Scraping Coordinator & Lead Developer: Support the one-time scraping workflow, validate extracted fields, manage duplicate entries, and document collection limitations.

2. Nourhan Ebrahim : Scraping & Data Validation Specialist: Support scraping across the three marketplaces, validate extracted fields, manage duplicate entries, and document collection limitations.

3. Marwa Mohamed : ETL & Data Cleaning Specialist: Clean and standardize fields, parse prices and specification text strings, handle missing values, and prepare the analysis-ready dataset.

4. Mennatollah Wael : Feature Engineering & DAX Specialist: Implement four price tiers, formulate Gaming, Camera, Battery, and Value-for-Money scores, and write DAX logic for cheapest-platform identification.

5. Kareem Elsayed : Power BI Dashboard & Analytics Lead: Build the six dashboard pages, KPI cards, visual charts, interactive slicers, navigation menus, and platform comparison views.

6. Mohamed Mahmoud : Insights, Recommendations & Presentation Specialist: Validate recommendation outputs, extract business insights, test interactions, and prepare the executive presentation deck and technical documentation.

3. Project Plan: Timeline, Milestones & Deliverables

Week 1: Project Setup & Data Collection

- Task Allocations:
    - Omar Adel: Define the scraping approach and build the initial extraction workflow.
    - Nourhan Ebrahim: Support the one-time collection and validate listings collected from Amazon Egypt, Noon Egypt, and Jumia Egypt.

    - Marwa Mohamed & Mennatollah Wael: Define the required schema, data dictionary, and cleaning rules.
    - Kareem Elsayed & Mohamed Mahmoud: Sketch dashboard wireframes and agree on key business questions and KPIs.
- Deliverables:
    - Initial raw datasets collected from Amazon Egypt, Noon Egypt, and Jumia Egypt, with the collection date/time documented.
    - Agreed data schema and comprehensive data dictionary.
    - Initial dashboard wireframe layouts and KPI list.
    - Week 1 progress review and issue log.

Week 2: Cleaning, Modeling & Feature Engineering

- Task Allocations:
    - Omar Adel & Nourhan Ebrahim: Review the collected files, consolidate the available records, and resolve data-quality issues where feasible; do not assume new scraping runs will occur.
    - Marwa Mohamed: Clean prices and specification fields, standardize brands/models, and handle missing values.
    - Mennatollah Wael: Implement Budget, Midrange, Upper Midrange, and Flagship price tiers.
    - Marwa Mohamed & Mennatollah Wael: Define and implement documented scoring logic for gaming, camera, battery, and value for money.
    - Kareem Elsayed & Mohamed Mahmoud: Review model fields and confirm dataset supports planned visuals and persona filters.
- Deliverables:
    - Cleaned, consolidated dataset.
    - Documented transformation and feature-engineering rules.
    - Initial score calculations and data-quality validation checks.
    - Draft Power BI relational data model.

Week 3: Dashboard Development & Recommendation Logic

- Task Allocations:
    - Kareem Elsayed: Build Market Overview, Platform Analytics, and Cross-Platform Price Matcher pages.
    - Mohamed Mahmoud: Build persona pages and the Interactive Deal Finder layout.
    - Mennatollah Wael: Implement and test DAX measures, cheapest-platform flags, and recommendation ranking formulas.
    - Marwa Mohamed: Verify cleaned fields, relationships, data types, and dashboard behavior when filtering the loaded static dataset.
    - Omar Adel & Nourhan Ebrahim: Investigate gaps in the existing scrape and correct source-data issues where feasible, using the collected data; any new scrape would require a separate manual run.
- Deliverables:
    - Working draft of the six-page Power BI dashboard.
    - Tested DAX measures and recommendation ranking logic.
    - Initial market and platform analytical findings.
    - Issue tracking list and prioritization for final validation.

Week 4: Validation, Storytelling & Final Submission

- Task Allocations:
    - Omar Adel & Nourhan Ebrahim: Document the one-time scraping methodology, collection date, source coverage, and limitations, including the fact that the dataset is not automatically updated.
    - Marwa Mohamed & Mennatollah Wael: Validate calculations, price tiers, score behavior, and cross-platform matches.
    - Kareem Elsayed & Mohamed Mahmoud: Refine dashboard visual design, write insight narratives, and prepare the executive presentation.
    - All Team Members: Conduct end-to-end testing using multiple budgets and user priorities, prepare the project demo, and practice answers for the evaluation board.

- Deliverables:
    - Final interactive Power BI dashboard (.pbix).
    - Executive presentation deck (.pptx).
    - Technical documentation covering data collection, cleaning, data model, scoring, and limitations.
    - Final validation checklist and project demonstration package.

4. Risk Assessment & Mitigation Plan

R-01: Scraping Blockers & DOM Structure Changes：
The e-commerce websites may change their HTML layout or block automated scrapers during the initial collection. Since the project uses a one-time scrape, missing or inaccessible records may remain absent from the static dataset.

R-02: String Parsing & Data Type Conversion Errors：
High
Text strings containing ratings (e.g., "3.1 out of 5") cause DataFormat.Error during Power Query numeric conversion.
Use Text.BeforeDelimiter in Power Query to safely extract digits before forcing data type changes.

R-03: Currency & Price Volatility：
Complex DAX measures may slow dashboard interactions when users apply slicers and budget filters.
Add a fixed Scrape_Date baseline column and document collection timeframes in the project limitations.

R-04: DAX Calculation & Measure Performance Slowdown：
Some smartphone listings in the collected dataset may lack critical fields such as battery capacity or refresh rate.
Leave unavailable values clearly marked as not specified; use a documented lookup only when the value can be verified. Avoid inventing missing specifications.

R-05: Missing Specification Data — Medium: Low
Certain smartphone listings lack critical fields like Battery mAh or Refresh Rate.
Impute missing specs using standardized brand lookup tables or explicitly label non-specified devices.

5. Key Performance Indicators (KPIs)

To evaluate the technical success, efficiency, and business effectiveness of the MobiGuide project, performance is measured against the following metrics:

Technical & Data Performance Metrics

- Collected-Data Completeness：Percentage of collected phone listings containing the required core fields (Name, Price, Brand, and Platform). Report the actual result from the dataset rather than treating the target as an achieved result.
- Data Transformation Quality (target: no unhandled errors): Check that the existing dataset can be cleaned and loaded without unhandled formatting errors.
- Dashboard Interaction Response Time (target: <2 seconds): Time required for Power BI visuals and recommendation tables to respond to slicers and budget selections using the loaded dataset.